DR. CHIA MIN YAN

Data Scientist | ML Engineer | Engineering PhD

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Loc: Selangor, Malaysia

PROFILE SUMMARY

Dr. Chia Min Yan is an engineering PhD and Data Scientist working at the intersection of Machine Learning, Generative AI, and process optimization. He has published 5+ research articles in top-tier Q1 Web of Science journals. He designs and implements production-grade predictive models, scalable microservice architectures, and autonomous agentic workflows that bridge scientific engineering with commercial AI applications.

DEVELOPMENT & CORE TOOLS

Python (ML)	PyTorch
TypeScript	JavaScript
Node.js	Fastify
FastAPI	Angular
SQL	PostgreSQL
MATLAB	

DEVELOPER TOOLS

Docker	Kubenetes
Git & CI/CD	LangChain
LLMs (RAG)	

PROFESSIONAL EXPERIENCE

Machine Learning Engineer
July 2026 – Present

Juice Up EV Grid Sdn. Bhd.

- Algorithms & Systems: Lead the end-to-end research, architecture, and deployment of production-grade statistical algorithms and systems (rewards program optimization, EV fleet charging schedules, station utilization) to drive EV charging network operations.
- Full Stack & Backend Architecture: Spearhead the technical design of scalable backend microservices, rewards database structures, and RESTful APIs, while directing frontend development of responsive EV fleet management portals and interactive customer dashboards.
- Generative AI Initiatives: Champion corporate AI strategies, orchestrating custom LLM integrations and advanced agentic frameworks to automate complex reasoning and EV customer support workflows.

Technologies: Python, TypeScript, Node.js, Angular, PostgreSQL, Git, Docker

Data Scientist
July 2022 – July 2026

PAIDChain Sdn. Bhd.

- Machine Learning Engineering: Research, build, and deploy core enterprise statistical models including real-time fraud detection systems, merchant risk anomaly trackers, system monitors, and algorithmic credit scoring structures.
- Full Stack Architecture: Lead the design and implementation of highly scalable backend RESTful APIs, database structures, and responsive web portals/interactive dashboards using Node.js, TypeScript, PostgreSQL, and Angular.
- Generative AI Initiatives: Champion corporate AI strategies by setting up, orchestrating, and launching custom LLM integrations, retrieval-augmented generation (RAG) models, and advanced agentic frameworks for workflow automation.

Technologies: Python, TypeScript, Node.js, Angular, PostgreSQL, Git, Docker, LangChain

Production Engineer
Jan 2019 – April 2019

Maxter Glove Manufacturing Sdn. Bhd.

- Monitored glove quality assurance workflows and parameters to satisfy international regulatory compliance and client product specifications.
- Standardized preventative line maintenance records and engineered key operations tracking documents to minimize system downtime.

Production Engineer (Intern)
Oct 2015 – Dec 2015

Maxter Glove Manufacturing Sdn. Bhd.

- Supervised and maintained treatment variables for the industrial wastewater facility, guaranteeing 100% compliance with national ecological parameters.
- Designed chemical dosing optimization strategies, successfully lowering operation costs and reducing chemical waste.

LANGUAGES

- English — Professional
- Chinese (Mandarin) — Native
- Bahasa Melayu — Professional
- Korean — Conversational

RESEARCH INTERESTS

- Environmental System Modelling
- Agentic Decision Architectures
- Applied ML in Process Systems
- Polymer composite membranes
- Sustainable fuel cells

PROFESSIONAL AFFILIATIONS

- Graduate Member — Board of Engineers Malaysia (BEM)
- Graduate Member — Institution of Engineers, Malaysia (IEM)
- Associate Member — Inst. of Chemical Engineers (IChemE)
- Graduate Technologist — Malaysia Board of Tech (MBOT)

EXTRACURRICULARS

- UTAR Chinese Debate Team (2018–2022)
  - Completed in the 16th and 17th National Varsity Chinese Debate Tournaments.
  - Represented UTAR in the 2018 Nan Ying Cup Debate in China.

EDUCATION

Doctor of Philosophy (Engineering) *2019 – 2022*

Universiti Tunku Abdul Rahman, Malaysia

Thesis: Robust Data Fusion Techniques Integrated Machine Learning Models for Estimating Reference Evapotranspiration. Developed deep learning models and custom stochastic optimization algorithms to resolve data sparsity.

Master of Engineering Science *2017 – 2019*

Universiti Tunku Abdul Rahman, Malaysia

Dissertation: Development of High Performance Sulfonated Poly (Ether Ether Ketone) Based Composite Proton Exchange Membranes. Investigated chemical parameters to increase proton fuel cell membrane performance.

Bachelor of Engineering (Honours) Chemical Engineering *2013 – 2016*

Universiti Tunku Abdul Rahman, Malaysia

CGPA: 3.8758 / 4.0000 (First Class Honours with Distinction)

SELECTED PUBLICATIONS

Wong, Y.J.*, Yeganeh, A., Chia, M.Y.*, et al. 2023. Quantification of COVID-19 impacts on NO2 and O3: Systematic model selection and hyperparameter optimization on AI-based meteorological-normalization methods. Atmospheric Environment. [Q1]

Wai, K.P., Chia, M.Y., Koo, C.H.*, Huang, Y.F. and Chong, W.C. 2022. Applications of deep learning in water quality management: A state-of-the-art review. Journal of Hydrology. [Q1]

Chia, M.Y., Huang, Y.F.*, Koo, C.H., Ng, J.L., Ahmed, A.N. and El-Shafie, A. 2022. Long-term forecasting of monthly mean reference evapotranspiration using deep neural network: A comparison of training strategies and approaches. Applied Soft Computing. [Q1]

Chia, M.Y., Huang, Y.F.* and Koo, C.H. 2022. Resolving data-hungry nature of machine learning reference evapotranspiration estimating models using inter-model ensembles with various data management schemes. Agricultural Water Management. [Q1]

Chia, M.Y., Huang, Y.F.* and Koo, C.H. 2021. Improving reference evapotranspiration estimation using novel inter-model ensemble approaches. Computers and Electronics in Agriculture. [Q1]

REFERENCES AVAILABLE UPON REQUEST